# Soil DNA extraction, manual (SuperSoil, legacy)

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Lab plain-PDF edition, rebuilt 2026-10-21

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**Abstract**

A method for fungal and bacterial DNA extraction.

**Safety warnings**

1. Solution SEA and Ethanol are flammable
2. DO NOT add bleach or acidic solutions directly to the sample preparation waste.

**Procedure**

### SAMPLE PREPARATION & CELL LYSIS

**1.** Add **0.2 mL** ⌀=1.0 mm zirconia beads, **0.2 mL** ⌀=0.5 mm zirconia beads, **800 µL** SD1 solution, **0.5 g** sample into a 1.5 ml microcentrifuge tube. Vortex briefly to mix.

**Note** If the SD1 solution is precipitated, leave it at room temperature for 30 minutes, it will become transparent again.

**Note** If the soil is too loose and fills up the whole microcentrifuge tube, you can separate them into 2 microcentrifuge tubes for the downstream workflow. And centrifuge 2 microcentrifuge tubes at the final step to merge them.

**Note** If your sample is hard to lyse, you can try the following methods:

1. After adding solution SD1, incubate at 65 °C for 10 minutes
2. Add 3 additional stainless steel beads and use 2.0 ml screw vials for your samples

**2.** Homogenize samples using Vortex-Genie® 2 with SI-H524 Horizontal microtube holder. Vortex at maximum speed for **4 min**

**Note** Be careful when removing the microcentrifuge tube from the holder. Press the lid and slowly loosen the tube with another hand.

**3.** Centrifuge the 1.5 ml microcentrifuge tube at **14000 rpm, 25 °C, 2 min**

### INHIBITOR PRECIPITATION

**4.** Add **200 µL** SD2 solution to a clean 1.5 ml microcentrifuge tube.

**5.** Transfer **500 µL** supernatant to the tube in the previous step. Vortex **5 s** for mixing.

**Note** White precipitate forms.

If too much AlCl3 (CD2) is added, it will cause all the DNA to be precipitated along with the inhibitory substances.

**Note** If there is any suspended substance, you can use P200 and turn it to 150 µl and suck the supernatant up 3 times.

**6.** Centrifuge at **14000 rpm, 25 °C, 2 min**

### DNA BINDING

**7.** Add **600 µL** SD3 solution to a clean 1.5 ml microcentrifuge tube. Avoiding the pellet, transfer up to **700 µL** of supernatant from the previous step to that tube. Vortex **5 s** for mixing.

**Note** If Solution CD3 has precipitated, heat at 60°C until the precipitate dissolves.

**8.** Take **650 µL** to spin column (with collection tube) and centrifuge at **14000 rpm, 25 °C, 2 min**

**9.** Discard the flow-through and repeat step 8 to ensure all the lysates have passed through the spin column.

**10.** Carefully place the spin column into a clean 2 ml collection tube. Avoid splashing any flow-through onto the spin column.

### WASH

**11.** Add **500 µL** of Solution SEA to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

**Note** Solution SEA may leaks or suck back causing contamination.

**12.** Discard the flow-through and return the spin column to the same 2 ml collection tube.

**13.** Add **500 µL** 80% ethanol to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

**Note** Ethanol may leaks or suck back causing contamination.

**14.** Discard the flow-through and place the spin column into a new 2 ml collection tube.

**15.** Centrifuge at **14000 rpm, 25 °C, 3 min** Carefully place the spin column into the new 1.5 ml microcentrifuge tube

**Note** Ethanol can interfere with downstream DNA applications, such as PCR, restriction digests, and gel electrophoresis. Make sure to centrifuge for 3 minutes.

### ELUTE

**16.** Add **50 µL** RNase-free water (DEPC-treated water) to the center of the white filter membrane.

**Note** Alternative for RNase-free water: TE(Tris-EDTA) buffer

**17.** Centrifuge at **14000 rpm, 25 °C, 2 min** Discard the spin column. The DNA will be in the microcentrifuge tube.

### QUALITY AND QUANTITY ASSESSMENT

**18.** Use BioTek Take3 & Take3 Trio Microvolume Plates to measure the quality and quantity of your specimens. And then upload them to Google Drive (1 MolecularWork (Direct edit) 分生實驗號碼 (直接編輯) w_ Primer list)

**Materials**

| # | Material |
|---|---|
| 1 | Bead (ball mill) |
| 2 | Solution SD1 (Chaotropic) |
| 3 | Solution SD2 |
| 4 | Solution SD3 (Chaotropic) |

| **Zirconia beads** | **0.5** | **6** | **0.2** | **CAAR-ZB-0.5-1000BX** |
|---|---|---|---|---|
| Zirconia beads | 1.0 | 6 | 0.2 | CAAR-ZB-1-1000BX |

| | **w/w** |
|---|---|
| sodium thiocyanate | 2.5% |
| $Na_2HPO_4$ | 5% |
| pH | ~8.4 |

| | **w/w** |
|---|---|
| $AlCl_3 \cdot 6H_2O$ | 1.5% |
| Ammonium acetate | 25% |
| pH (Adjust with acetic acid) | 6.3 |

| | **w/w** |
|---|---|
| Guanidinium thiocyanate | 65% |
| pH | ~7.0 |

Solution SEA (Wash buffer 1)

| | **w/w** |
|---|---|
| Ethanol | 50% |
| Isopropanol | 10% |
| Guanidine hydrochloride | 30% |

80% Ethanol (Wash buffer 2)

DEPC treated water (Elution buffer)